like a glowing poker. But black holes, as everyone knows, are black; they supposedly do not emit anything. Hawking and most everyone else agreed that this definitively ruled out Bekenstein's suggestion. Instead, Hawking was willing to accept that if matter carrying entropy is dropped into a black hole, this entropy is lost, plain and simple. So much for the second law of thermodynamics.

This was the case until Hawking, in 1974, discovered something truly amazing. Black holes, Hawking announced, are not completely black. If one ignores quantum mechanics and invokes only the laws of classical general relativity, then as originally found some six decades previously, black holes certainly do not allow anything—not even light—to escape their gravitational grip. But the inclusion of quantum mechanics modifies this conclusion in a profound way. Even though he was not in possession of a quantum-mechanical version of general relativity, Hawking was able to finesse a partial union of these two theoretical tools that yielded some limited yet reliable results. And the most important result he found was that black holes do emit radiation, quantum mechanically.

The calculations are long and arduous, but Hawking's basic idea is simple. We have seen that the uncertainty principle ensures that even the vacuum of empty space is a teeming, roiling frenzy of virtual particles momentarily erupting into existence and subsequently annihilating one another. This jittery quantum behavior also occurs in the region of space just outside the event horizon of a black hole. Hawking realized, however, that the gravitational might of the black hole can inject energy into a pair of virtual photons, say, that tears them just far enough apart so that one gets sucked into the hole. With its partner having disappeared into the abyss of the hole, the other photon of the pair no longer has a partner with which to annihilate. Instead, Hawking showed that the remaining photon gets an energy boost from the gravitational force of the black hole and, as its partner falls inward, it gets shot outward, away from the black hole. Hawking realized that to someone looking at the black hole from the safety of afar, the combined effect of this tearing apart of virtual photon pairs, happening over and over again all around the horizon of the black hole, will appear as a steady stream of outgoing radiation. Black holes glow.